Research Protocol: Evaluation of Usability and Experience in Configuring AWS Architectures

1. Session Objective

The core purpose of this session is to collect qualitative and perception data on the usability, effectiveness, and user experience of interacting with random combinations of AWS services. Although the activity is presented to students as a technical problem-solving challenge to validate their architecture skills, the underlying objective is to evaluate how different service combinations affect cognitive load and the appeal of digital game-based learning.

2. Step-by-Step Procedure

A. Introduction and Framing

The session begins by explaining to students that they will take part in a cloud technical competency validation test. They are told that the system will generate infrastructure scenarios based on real cases to assess their responsiveness and architectural design ability.

B. Group and Service Assignment

Participants are divided into 4 to 5 groups. Each group is assigned a set of AWS services randomly generated by the program, ensuring there are no incompatible combinations so that the scenarios reflect architectures possible in production environments.

C. Survey Administration

Each student must individually complete a structured questionnaire covering two main dimensions:

- **Block A (Technical Dimension):** Focus on problem resolution. Questions about the connection logic, deployment, and functional relationship between the assigned services.

- **Block B (Perception Dimension):** Evaluation of the impact of digital game-based learning, measuring usability, perceived effectiveness, interface appeal, and user experience.

D. Focus Group Session

After the technical work is finished, the groups are separated for a semi-structured interview. The goal is to explore aspects the questionnaire does not capture.

3. Survey Questions

Block A:

- Resolution Question 1: Describe how you would integrate the assigned services into a functional solution and what the data flow between them would be.
- Resolution Question 2: In which real-world scenarios would you consider this combination of services applicable?

Block B *(1 = Strongly disagree, 6 = Strongly agree):*

- The format of the challenge allows me to clearly visualize the interaction of AWS services in a real environment.
- Solving random infrastructures makes it easier to understand complex cloud concepts.
- The dynamic of the exercise is more engaging than a traditional guided lab.
- The rules and objective of the technical challenge are easy to understand.
- The integration of the assigned services is unnecessarily complex.
- The level of ambiguity of the assigned services makes the exercise harder to solve.
- Expert support or extensive documentation is required to deploy this scenario.
- The conditions of the exercise allow me to apply technical judgment intuitively and confidently.

4. Focus Group Questions

- **Decision-making under uncertainty:** When the system assigned you a combination of services that seemed unusual or "weird," what was your initial reaction, and how did you mentally structure your first steps to make sense of the problem?

- **Frustration tolerance:** Was there a point in the design where you felt the architecture simply would not fit or was redundant? How did you handle that technical barrier so as not to get stuck in the exercise?
- **Adaptability:** If you had to defend this exact architecture in front of a real client, what arguments (technical, cost-related, or business-related) would you use to justify your decisions under pressure?
- **Cognitive self-regulation:** Comparing this activity with traditional cloud exercises you have done before, how would you describe the mental effort and stress management required to take on the role of architect facing an open-ended problem?

Base References

- Barana, A. (2025). DOI: 10.1109/compsac65507.2025.00333.
- Arif, Y. M. (2026). DOI: 10.1109/ACCESS.2026.3669416